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fastfrechet: An R package for fast implementation of Fréchet regression with distributional responses

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Summary

Distribution-as-response regression problems are gaining wider attention, especially within biomedical settings where observation-rich patient specific data sets are available, such as feature densities in CT scans (Petersen et al., 2021), actigraphy (Ghosal et al., 2023), and continuous glucose monitoring (Coulter et al., 2024; Matabuena et al., 2021). To accommodate the complex structure of such problems, Petersen & Müller (2019) proposed a regression framework called *Fréchet regression* which allows non-Euclidean responses, including distributional responses. This regression framework was further extended for variable selection by Tucker et al. (2023), and Coulter et al. (2024) developed a fast variable selection algorithm for the specific setting of univariate distributional responses equipped with the 2-Wasserstein metric (*2-Wasserstein space*). We present `fastfrechet`, an R package providing fast implementation of these Fréchet regression and variable selection methods in 2-Wasserstein space, with resampling tools for automatic variable selection. `fastfrechet` makes distribution-based Fréchet regression with resampling-supplemented variable selection readily available and highly scalable to large data sets, such as the UK Biobank (Doherty et al., 2017).

Statement of Need

Fréchet regression with variable selection is currently not implemented by any software package, available only through the Supplementary Material of Tucker et al. (2023) (hereafter "`Tucker` materials"). As discussed in Coulter et al. (2024), the implemented algorithm is prohibitively slow. Implementation of Fréchet regression in 2-Wasserstein space without variable selection is supported by the `Tucker` materials, and by two R packages: `WRI` (Liu et al., 2022) and `frechet` (Chen et al., 2023). These packages face certain practical limitations. For instance, `WRI` requires continuous distributions, and does not allow user-specified constraints for the distribution support. `frechet` offers more flexibility in user specifications, but its solver for Fréchet regression is slow and may not accurately satisfy constraints.

The `fastfrechet` package addresses these limitations by providing a fast, scalable, and user-friendly implementation of both Fréchet regression and variable selection for 2-Wasserstein space, based on the work of Coulter et al. (2024). The package incorporates resampling tools to enhance automatic variable selection, including cross-validation described in Tucker et al. (2023) and stability selection described in Coulter et al. (2024). Additionally, `fastfrechet` features a customized dual active-set solver for the Fréchet regression problem, inspired by Arnström et al. (2022), which ensures both computational efficiency and accuracy while accommodating user-specified support constraints. The Fréchet regression solver also accommodates the auxiliary weighting scheme used in the variable selection procedure, the first package to do so.

Performance Comparisons to Existing Implementations

We illustrate the performance of `fastfrechet` against existing implementations with simulated covariate-dependent distributional responses. The included function `generate_zinbinom_qf` simulates n zero-inflated negative binomial (**zinbinom**) distributions (we choose $n = 100$), represented as quantile functions evaluated on a shared m-grid in $(0, 1)$ (we choose $m = 100$), and dependent on the first 4 of $p \geq 4$ covariates (we choose $p = 10$). We utilize the R package `microbenchmark` (Mersmann, 2024) to calculate run times, and report median times for each method (Fréchet regression, variable selection) from 15 iterations; all computations were performed on an Apple M1 Max chip. To replicate the specific simulation and comparison settings used in this manuscript, see the accompanying `performanceExample-fastfrechet` vignette.

The Fréchet Regression Problem

`fastfrechet` provides a solver for the Fréchet regression problem for 2-Wasserstein space, with optional `lower` and `upper` support constraints on the underlying distributions. Since **zinbinom** distributions are non-negative, we fix `lower = 0` and `upper = Inf` (or some suitably large number, as applicable). The regression outputs are fitted quantile functions, which should be monotone non-decreasing and obey support constraints. The `fastfrechet` implementation is a customization of the dual active-set method of Arnström et al. (2022). (See the accompanying `monotoneQP-fastfrechet` vignette for full algorithm description.)

Figure 1 illustrates the speed and accuracy of Fréchet regression implemented in `fastfrechet` against the `WRI`, `frechet`, and `Tucker materials` implementations. `WRI` does not accept known support bounds as input, and fitted responses correspondingly violate the zero lower bound; `frechet` solutions only approximately satisfy the lower bound. The `Tucker materials` implementation finds numerically accurate solutions, but `fastfrechet` accomplishes this in a fraction the time.

The Variable Selection Problem

The R package `fastfrechet` implements variable selection for Fréchet regression, specifically in 2-Wasserstein space. Variable selection comprises finding optimal weight vector $\hat{\boldsymbol{\lambda}} \in \mathbb{R}^p$ that satisfies a τ-simplex constraint, given hyperparameter $\tau > 0$. In 2-Wasserstein space, $\hat{\boldsymbol{\lambda}}$ essentially minimizes an L^2 norm between weighted Fréchet regression outputs $\widehat{\boldsymbol{Q}}(\hat{\boldsymbol{\lambda}})$ and the raw data $\boldsymbol{Y}$. (See the accompanying `intro-fastfrechet` vignette for a detailed exposition.) `fastfrechet` implements the second-order geodesic descent algorithm developed by Coulter et al. (2024), with two modifications. First, the implementation uses the custom dual active-set method discussed in the previous subsection. The active set defining the weighted Fréchet regression solution $\widehat{\boldsymbol{Q}}(\boldsymbol{\lambda}^t)$ for iterate $\boldsymbol{\lambda}^t$ serves as a warm start for iterate $\boldsymbol{\lambda}^{t+1}$, reducing computation time. Second, the implementation allows the user to specify an impulse parameter, which implements momentum-based geodesic descent.

Figure 2 illustrates the speed and accuracy of variable selection implemented in `fastfrechet` against the `Tucker materials` implementation, across sequence of hyperparameter values $\tau \in \{0.5, 1.0, \cdots, 10.0\}$. We hand-select `fastfrechet` error tolerance parameter $\varepsilon = 0.014$, which gives solutions $\hat{\boldsymbol{\lambda}}(\tau)$ minimizing the objective function approximately as well as solutions from the other method "as-is". `fastfrechet` is upward of 20,000$\times$ faster to obtain these comparable solutions. Decreasing the `fastfrechet` error tolerance parameter increases optimization accuracy with modest increases in computation time.

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Figures

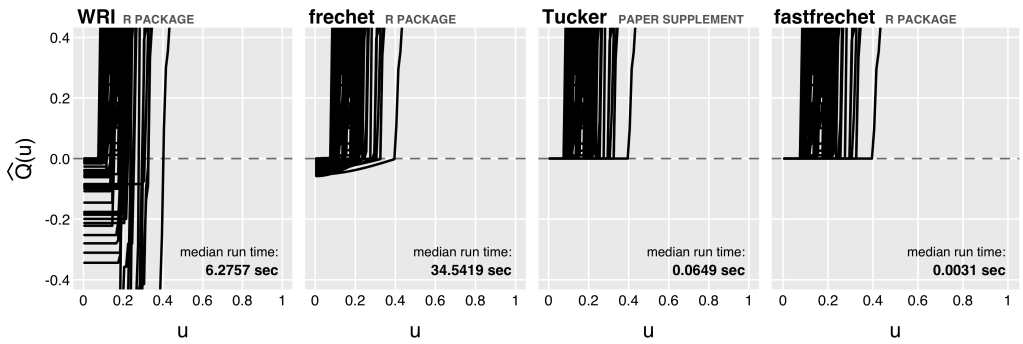

Figure 1: Fitted Fréchet regression quantile functions (zoomed in around zero) and median run times for `fastfrechet` and other implementations. Fitted quantile functions below zero violate known lower support constraints.

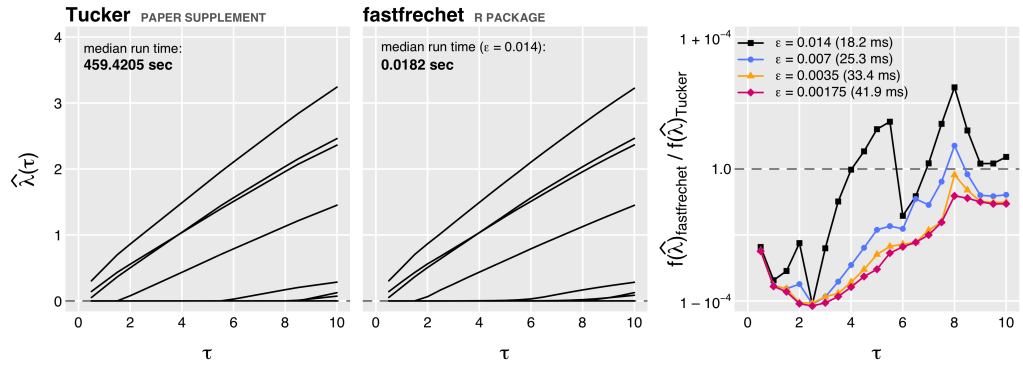

Figure 2: (**left, center**) Variable selection solution paths $\hat{\boldsymbol{\lambda}}(\tau)$ across $\tau = \{0.5, 1, \ldots, 10\}$ and median run times for `Tucker` materials and `fastfrechet`. (**right**) Relative optimization accuracy of `fastfrechet` and `Tucker` materials variable selection, and median `fastfrechet` run times, using different error tolerance values. Points below 1.0 indicate `fastfrechet` solutions minimize the objective function better.

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